

Zane Wang

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Summary

Results-oriented full-stack engineer with 5+ years architecting and shipping scalable production systems across Node.js, Python, TypeScript, and React.js, backed by strong fundamentals in distributed data (MongoDB, PostgreSQL) and cloud infrastructure (GCP, Azure, AWS). Experienced in AI-augmented development — integrating LLM/agent workflows (Claude Code with a custom ruleset) into daily engineering to accelerate delivery and code quality. Proven track record of owning services end-to-end, from technical design through deployment, for products used by thousands, and driving pragmatic technical decisions in Agile teams.

Projects

UQ Ask Anything

Built a retrieval-augmented QA service answering UQ students' questions on courses, prerequisites, fees, and policy, over Postgres + pg vector. An LLM planner routes each question to structured, semantic, or hybrid retrieval, while all high-risk enrolment facts are answered by deterministic code — zero hallucination — with the LLM confined to drafting grounded, citation-guarded wording.

- Architected a planner-routed agentic RAG pipeline with six retrieval modes (filter / semantic / hybrid / program / course-detail / KB), combining SQL filtering, bge-m3 semantic search, and RRF hybrid retrieval.
- Designed the planner to fill only validated filter slots and extract entities, with SQL assembled from bound parameters so the model never writes raw queries.
- Enforced a strict deterministic-vs-LLM boundary so factual data (prerequisites, fees, enrolment) comes from code, plus a KB answerability gate that refuses to answer when evidence is missing.
- Built a pluggable LLM backend (Ollama / DeepSeek / Bedrock) for flexible model routing.

Work Experience

Broadsheet, Melbourne, Australia

Full Stack Developer | Feb 2025 - Jul 2026

Broadsheet Media Website

Technologies: Next.js, React.js, TypeScript, Node.js, Python, PostgreSQL, Redis, GCP, Docker, Kubernetes, CI/CD, Jest

- Architected and shipped responsive, scalable web applications with Next.js, TypeScript, and Node.js, integrating RESTful APIs and Redis caching to drive measurable performance and reliability gains
- Designed an event-driven backend on GCP (Cloud Run, GKE, Pub/Sub) with message-queue-based event-bus architecture and PostgreSQL schema design, ensuring high availability and horizontal scalability
- Owned the full delivery pipeline—containerised CI/CD with Docker and Kubernetes plus Jest/Python test coverage—embedding quality and deployment automation across microservices
- Pioneered AI-augmented development by integrating Claude Code, Codex, and Cursor into the team workflow, accelerating delivery velocity without compromising code quality

- Drove engineering standards through code review and cross-functional Agile collaboration, aligning technical decisions with product and user needs

AnyStay, Brisbane, Australia

Full Stack Developer | Dec 2023 – Dec 2024

Description: Online innovative e-commerce platform for short-term and long-term stays, including hosts and guests system, sales, member portal and membership management.

Anystay Website

Technologies: React.js, Redux, Node.js, Python, GraphQL, DynamoDB, AWS, CI/CD, Jest, Selenium

- Built responsive web applications with React.js (Redux) and GraphQL, backed by RESTful APIs in Node.js, delivering performant and maintainable full-stack solutions
- Designed DynamoDB table structures and access patterns—optimising partition/sort keys and indexes—to support high-throughput, low-latency read/write workloads at scale
- Architected an event-driven service bus on a message queue to decouple core systems, and shipped them via CI/CD to AWS Amplify (frontend) and Lambda (backend)
- Drove quality through Jest unit tests, Python Selenium automation, and comprehensive code reviews that strengthened design patterns and system architecture
- Led projects in a fast-paced Agile environment, partnering with UX/UI and cross-functional teams to consistently meet deadlines and ship outstanding user experiences

ByteDance, Beijing, China

Full Stack Developer | Jul 2023 – Dec 2023

Description: Leading tech company that leverages data and AI to innovate social media platforms and enhance user engagement through content discovery.

Technologies: React.js, TypeScript, Node.js (Express.js), Python, PostgreSQL, Redis, Kafka, Docker, Kubernetes, Terraform, CI/CD, Jest

- Built full-stack web middleware with React.js, TypeScript, and Node.js (Express.js) over PostgreSQL, designing schemas and tuning indexes/queries for reliable read/write performance across services
- Implemented a Redis-backed queue system that cut middleware latency by 60%, and introduced event-driven architecture via message queue to decouple core services and scale under high-throughput workloads
- Containerised middleware and API services with Docker and orchestrated deployments on Kubernetes, ensuring consistent runtime across dev/staging/prod environments
- Built and maintained CI/CD pipelines for frontend and backend, with Terraform managing infrastructure as code for reproducible environments
- Safeguarded quality through automated API and unit tests (Python, Jest) while consistently delivering in a fast-paced Agile environment

Kexing EasyGo, Beijing, China
Full Stack Developer | Jan 2022 - Jul 2023

Description: Designing mobile applications for University of Queensland students that consolidates essential academic tools into a user-friendly platform for managing grades, deadlines, schedules, and course evaluations.

Technologies: React.js, React Native, TypeScript, TailwindCSS, Redux, Node.js, PostgreSQL, MongoDB, Kafka, AWS, Docker, Kubernetes, Terraform, CI/CD, Jest, Selenium, Postman

- Built full-stack applications with React.js, Tailwind CSS, TypeScript, and Redux on the frontend, backed by RESTful APIs in Node.js/TypeScript
- Architected a Kafka-based event-bus for pub/sub messaging across microservices, enabling durable event processing and reducing tight coupling between backend components
- Containerized services with Docker and orchestrated them on Kubernetes (AWS) for reliable horizontal scaling, with CI/CD pipelines and Terraform-provisioned infrastructure enabling repeatable, automated deployments
- Established a Jest unit-testing standard across frontend and backend, integrated into CI with enforced coverage thresholds to catch regressions early
- Led a team of frontend and backend developers, driving code reviews and providing technical support

Graviti, Shanghai, China
Full Stack Developer | Jan 2021 – Jan 2022

Description: Graviti is a technology company dedicated to transforming AI development by providing next-generation tools for efficient dataset acquisition, storage, and processing, empowering industries with artificial intelligence.

Technologies: React.js, TypeScript, Python, Golang, MongoDB, RabbitMQ, Docker, Kubernetes, CI/CD, Jest

- Designed and built a React.js/TypeScript dashboard alongside a distributed web-crawling system in Python, with custom API Gateway and RESTful APIs in Golang
- Architected MongoDB document schemas with indexing strategies for crawled datasets, supporting efficient querying across large volumes of unstructured data
- Built an asynchronous RabbitMQ job queue to coordinate distributed crawlers and downstream processing, decoupling crawl scheduling from data ingestion
- Packaged services with Docker and deployed on Kubernetes for elastic scaling of crawler workers under variable load, with unit tests integrated into CI to enforce coverage before merge
- Delivered within 2-week Agile sprints, partnering with UX/UI designers, product managers, and cross-functional teams to ship exceptional user-centered designs

Skills

Languages: Python, JavaScript, TypeScript, Golang
Web tech: Jest, RESTful API, Redux
Front end: HTML5, CSS3, jQuery, Next.js, React.js, React Native, Node.js, Typescript, Bootstrap, TailwindCSS
DB: MySQL, PostgreSQL, MongoDB, DynamoDB
Cloud: GCP, Docker, Kubernetes, CI/CD AWS, GitHub Actions, Jenkins, Terraform
Messaging: RabbitMQ, Redis
Tools: Expo, Git, Jenkins, Jira, Figma, Postman

Education

- Master of Information Technology (Artificial Intelligence)
University of Melbourne, Australia
- Bachelor of Engineering (Electrical Engineering)
University of Queensland, Australia

Referees

Available upon requests